Finite Readout Certificates for Navier–Stokes: Discrete Epsilon-Completion, Spectral Tails, and A Posteriori Adapters

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Abstract

Finite numerical agreement is not, by itself, a certificate that an infinite-dimensional Navier–Stokes state has been reconstructed to a declared tolerance. This paper develops a fail-closed formulation of that distinction. A nested finite-refinement defect δ_K measures disagreement on shared retained information, while a separate quantity $\beta_{K,Y}$ is required to bound distinctions omitted by the finite representation in a declared target space Y. We first prove an elementary obstruction: terminal Fourier coefficients below a cutoff cannot, over an unrestricted divergence-free L^2 class, determine a universal useful bound on the omitted terminal tail. We then identify target spaces and retained records for which rigorous bounds do exist. For an unforced Leray–Hopf solution on the periodic three-torus,

$$\|(I - P_K)u\|_{L^2(0,T;L^2_x)} \leq \frac{\|u_0\|_2}{\sqrt{2\nu}\,(K+1)},$$

which yields a computable $O(K^{-1})$ spacetime certificate without assuming global smoothness. A terminal L^2 certificate follows either from a separately certified pointwise H^s bound or from a richer retained energy/dissipation tape. Finally, a residual-based relative-energy estimate connects a smooth finite Fourier comparison path to a Leray–Hopf solution and isolates the remaining numerical obligation: a validated continuous-time enclosure of the actual RK4 tape. The associated finite Fourier residual is itself a finite triad computation. Executable finite diagnostics and a cross-repository provenance ledger accompany the analysis. None of the results proves global regularity, finite-time blow-up, uniqueness of weak solutions, or any part of the Clay Millennium Navier–Stokes problem.

1 Introduction

A finite computation answers a finite question. A continuum claim requires an additional bridge. This elementary observation becomes operationally important in numerical PDE work, where several distinct statements are easily conflated:

$$\text{finite algebra correct} \quad \not\Rightarrow \quad \text{continuum complete} \quad \not\Rightarrow \quad \text{physically validated turbulence.}$$

The present paper studies how the middle implication can be made explicit and fail-closed for incompressible Navier–Stokes computations.

The starting point is the Discrete Epsilon-Completion (EPSC) programme. At refinement level K, a finite state or readout x_K is compared with the restriction of the next refinement. Small nested disagreement is useful evidence, but it does not control distinctions that both finite representations omit. We therefore separate two quantities:

$$\delta_K = \|R_K x_{K+1} - x_K\|, \tag{1}$$
$$\beta_{K,Y} = \text{a proved upper bound on omitted information in target } Y. \tag{2}$$

Only when the relevant hypotheses are certified does the fail-closed gate permit

$$\delta_K + \beta_{K,Y} \le \varepsilon \quad \Longrightarrow \quad \mathrm{CERTIFIED}_\varepsilon(Y).$$

If the required $\beta_{K,Y}$ is unavailable, the verdict is `HOLD` regardless of how small δ_K appears numerically.

The contribution here is not a new energy inequality or a new weak–strong stability theorem. Those analytic ingredients are classical. The contribution is the explicit separation of target, retained record, adapter, and omitted-information certificate, together with a concrete Fourier implementation path and machine-executable finite checks. This separation identifies precisely which parts of the finite-to-continuum claim are already certified and which remain open.

2 Setting and notation

Let u solve incompressible Navier–Stokes on the 2π-periodic three-torus $\mathbb{T}^3$,

$$\partial_t u + \mathbb{P}[(u \cdot \nabla)u] - \nu \Delta u = \mathbb{P} f, \qquad \nabla \cdot u = 0, \tag{3}$$

with viscosity $\nu > 0$, where $\mathbb{P}$ is the Leray projector. We use the Fourier normalization

$$\|u\|_2^2 = \sum_{k \in \mathbb{Z}^3} |\widehat{u}_k|^2, \qquad \|\nabla u\|_2^2 = \sum_{k \in \mathbb{Z}^3} |k|^2 |\widehat{u}_k|^2.$$

Let P_K retain the cube $\|k\|_\infty \le K$ and write $Q_K = I - P_K$. Every omitted mode then satisfies $|k|_2 \ge K + 1$.

For the finite computation, the production representation is a set of integer mode labels and complex divergence-free coefficients. A finite Fourier–Galerkin trajectory obeys

$$\frac{d\widehat{v}_k}{dt} = -\nu |k|^2 \widehat{v}_k - i P_k \sum_{p+q=k} (q \cdot \widehat{v}_p) \widehat{v}_q, \qquad \|k\|_\infty \le K, \tag{4}$$

with all sums finite after the cutoff is fixed. The implementation used in the accompanying repository evaluates these ordered triads directly and does not require a spatial grid in the production path. This is a computational statement about the finite recurrence, not an ontological statement that continuum space has been eliminated.

3 Discrete epsilon-completion

Definition 1 (Nested consistency defect)**.** Given a restriction map R_K from refinement $K + 1$ to the information represented at level K, define

$$\delta_K(x) := \|R_K x_{K+1} - x_K\|. \tag{5}$$

For Fourier Navier–Stokes diagnostics, one may also record the outer retained-shell energy

$$E_{\partial K}(u) = \frac{1}{2} \sum_{\|k\|_\infty = K} |\widehat{u}_k|^2. \tag{6}$$

This is a finite diagnostic proxy for unresolved activity. It is not a theorem-level substitute for a bound on the full omitted tail.

Definition 2 (Target-indexed omitted-information certificate)**.** Let Y be a declared target norm or readout space and $\mathcal{R}_K$ a declared finite retained record under assumptions $\mathcal{A}$. An EPSC tail certificate is a computable quantity $\beta_{K,Y}(\mathcal{R}_K; \mathcal{A}) \geq 0$ such that

$$\|Q_K x\|_Y \leq \beta_{K,Y}(\mathcal{R}_K; \mathcal{A}). \tag{7}$$

For asymptotic completion one additionally requires $\beta_{K,Y} \to 0$ along the declared refinement sequence.

This definition makes explicit that "complete" is incomplete language unless the reader/norm, admissible class, and available retained record are specified.

4 Terminal finite-record obstruction

Proposition 3 (Low terminal modes do not identify an unrestricted tail)**.** *Fix $K \geq 0$. Over the class of divergence-free, real-valued L^2 Fourier states, the retained terminal record $P_K u$ alone does not determine any finite universal upper bound on $\|Q_K u\|_2$.*

Proof. Choose $q = (K+1, 0, 0)$ and a nonzero vector $a \perp q$. Define a real divergence-free perturbation v by placing Fourier coefficients only at q and $-q$, with $\widehat{v}_q = a$ and $\widehat{v}_{-q} = \overline{a}$. Then $P_K v = 0$, while $\|Q_K v\|_2 = \sqrt{2}|a|$. Scaling a leaves the retained record unchanged and makes the omitted tail arbitrarily large. Hence no function of $P_K u$ alone can give a finite universal terminal bound on the unrestricted class. $\square$

Remark 4. Proposition 3 is an identifiability statement about a terminal state class. It does not assert that Navier–Stokes dynamics realizes every such perturbation at a prescribed time. Rather, it shows why a useful terminal certificate must contain additional admissible-class or dynamical information.

5 A positive spacetime certificate

Lemma 5 (Spectral tail inequality)**.** *For $s > 0$ and any u with finite homogeneous H^s seminorm,*

$$\|Q_K u\|_2 \leq \frac{\||\nabla|^s u\|_2}{(K+1)^s}. \tag{8}$$

Proof. Every omitted mode has $|k| \geq K + 1$. Therefore

$$\sum_{\|k\|_\infty > K} |\widehat{u}_k|^2 \leq (K+1)^{-2s} \sum_{\|k\|_\infty > K} |k|^{2s} |\widehat{u}_k|^2,$$

and taking square roots gives the result. $\square$

Theorem 6 (Unforced Leray–Hopf spacetime tail certificate)**.** *Let u be a Leray–Hopf weak solution of the unforced equation on $[0, T]$. Then*

$$\boxed{\|Q_K u\|_{L^2(0,T;L^2_x)} \le \beta_K^{\mathrm{ST}} := \frac{\|u_0\|_2}{\sqrt{2\nu}\,(K+1)}} \tag{9}$$

and $\beta_K^{\mathrm{ST}} \to 0$ as $K \to \infty$.

Proof. Lemma 5 with $s = 1$ gives

$$\|Q_K u(t)\|_2^2 \le (K+1)^{-2}\|\nabla u(t)\|_2^2.$$

The Leray–Hopf energy inequality gives

$$\frac{1}{2}\|u(t)\|_2^2 + \nu\int_0^t \|\nabla u(s)\|_2^2\,ds \le \frac{1}{2}\|u_0\|_2^2.$$

Integrating the spectral bound and discarding the nonnegative terminal kinetic energy yields

$$\int_0^T \|Q_K u\|_2^2 dt \le \frac{\|u_0\|_2^2}{2\nu(K+1)^2}.$$

Taking square roots proves (9). ☐

For forcing $f \in L^2(0, T; H^{-1})$, Young's inequality gives the conservative variant

$$\|Q_K u\|_{L^2_t L^2_x} \le \frac{1}{K+1}\left(\frac{\|u_0\|_2^2}{\nu} + \frac{\|f\|_{L^2_t H^{-1}}^2}{\nu^2}\right)^{1/2}. \tag{10}$$

Corollary 7 (Lipschitz readout lift)**.** *If a declared readout $\mathcal{Q} : Y \to Z$ is $L_{\mathcal{Q}}$-Lipschitz in a space Y for which an EPSC certificate $\|u - P_K u\|_Y \le \beta_{K,Y}$ has been established, then*

$$\|\mathcal{Q}(u) - \mathcal{Q}(P_K u)\|_Z \le L_{\mathcal{Q}}\beta_{K,Y}.$$

For example, for the time average $\bar{u} = T^{-1}\int_0^T u(t)dt$, Cauchy–Schwarz gives

$$\|Q_K \bar{u}\|_2 \le \beta_K^{\mathrm{ST}}/\sqrt{T}.$$

6 Terminal certificates from richer retained records

6.1 A conditional H^s certificate

If a separate argument certifies

$$\||\nabla|^s u(T)\|_2 \le M_s(T),$$

then Lemma 5 immediately gives

$$\boxed{\|Q_K u(T)\|_2 \le \frac{M_s(T)}{(K+1)^s}.} \tag{11}$$

The difficult part in three dimensions is not the spectral inequality but obtaining an appropriate globally valid pointwise regularity bound. EPSC therefore does not bypass the classical regularity obstacle for arbitrary terminal reconstruction.

6.2 An energy/dissipation tape

The terminal no-go concerns low terminal coefficients alone. The energy inequality provides a natural richer retained record.

Theorem 8 (Terminal energy-budget certificate)**.** *Let u be an unforced Leray–Hopf solution. Suppose finite certified data satisfy*

$$U_0 \geq \|u_0\|_2, \qquad L_K(T) \leq \|P_K u(T)\|_2,$$

and

$$D_K(T) \leq \nu \int_0^T \|\nabla P_K u(t)\|_2^2 dt.$$

Then, whenever the right-hand side is nonnegative,

$$\boxed{\|Q_K u(T)\|_2 \leq \beta_K^{\mathrm{EB}} := \left[U_0^2 - L_K(T)^2 - 2D_K(T)\right]^{1/2}.} \tag{12}$$

A materially negative bracket signals inconsistent directional certificates and must produce a fail-closed verdict rather than being clamped to zero.

Proof. Orthogonality of Fourier projection gives

$$\|u(T)\|_2^2 = \|P_K u(T)\|_2^2 + \|Q_K u(T)\|_2^2,$$

and likewise

$$\|\nabla u\|_2^2 = \|\nabla P_K u\|_2^2 + \|\nabla Q_K u\|_2^2.$$

Insert both decompositions into the energy inequality and discard the nonnegative unresolved dissipation term. Replacing exact quantities by bounds in the indicated directions yields (12). □

With exact projected quantities, define the energy-inequality slack

$$\mathcal{D}_E(T) := \|u_0\|_2^2 - \|u(T)\|_2^2 - 2\nu \int_0^T \|\nabla u\|_2^2 dt \geq 0. \tag{13}$$

Then the exact squared budget decomposes as

$$(\beta_K^{\mathrm{EB}})^2 = \|Q_K u(T)\|_2^2 + 2\nu \int_0^T \|\nabla Q_K u\|_2^2 dt + \mathcal{D}_E(T). \tag{14}$$

Consequently

$$\lim_{K\to\infty} (\beta_K^{\mathrm{EB}})^2 = \mathcal{D}_E(T).$$

If energy equality holds, $\mathcal{D}_E(T) = 0$ and this particular terminal certificate tends to zero. For a general Leray–Hopf solution the actual Fourier tail still tends to zero at each fixed L^2 time, but the energy-budget certificate can retain a nonzero floor because the energy inequality alone does not account for the slack.

7 A residual-based Galerkin-to-continuum adapter

Theorem 8 concerns the actual continuum projection. A numerical Galerkin trajectory must not be silently substituted for $P_K u$. A standard relative-energy estimate gives an explicit adapter.

Let v be a smooth divergence-free comparison path and define its full-equation residual

$$r := \partial_t v + \mathbb{P}[(v \cdot \nabla)v] - \nu \Delta v - \mathbb{P} f. \tag{15}$$

Let $e_0 \geq \|u_0 - v(0)\|_2$ and define

$$A_T := 2 \int_0^T \|\nabla v(t)\|_\infty dt, \qquad B_T := \int_0^T \|r(t)\|_{H^{-1}}^2 dt. \tag{16}$$

Theorem 9 (Relative-energy adapter)**.** *Let u be a Leray–Hopf weak solution of* (3) *and v a sufficiently smooth, divergence-free comparison path with the same boundary conditions and forcing convention. Then*

$$\boxed{\sup_{0 \leq t \leq T} \|u(t) - v(t)\|_2^2 \leq e^{A_T} \left(e_0^2 + \frac{B_T}{\nu} \right).} \tag{17}$$

Proof. Set $w = u - v$. Testing the difference equation with w and using the incompressible cancellation leaves the deformation term $\langle (w \cdot \nabla)v, w \rangle$. Therefore

$$\frac{1}{2} \frac{d}{dt} \|w\|_2^2 + \nu \|\nabla w\|_2^2 \leq \|\nabla v\|_\infty \|w\|_2^2 + \|r\|_{H^{-1}} \|\nabla w\|_2.$$

Young's inequality absorbs half the viscous term and gives

$$\frac{d}{dt} \|w\|_2^2 \leq 2 \|\nabla v\|_\infty \|w\|_2^2 + \nu^{-1} \|r\|_{H^{-1}}^2.$$

Gronwall's inequality and the definitions (16) prove (17). □

If $v(T)$ is supported in P_K, then $Q_K v(T) = 0$, so

$$\boxed{\|Q_K u(T)\|_2 \leq \beta_K^{\mathrm{RE}} := e^{A_T/2} \left(e_0^2 + \frac{B_T}{\nu} \right)^{1/2}.} \tag{18}$$

This is a terminal continuum bound obtained from a comparison path and certified residual summary; it does not require identifying the numerical state with the continuum projection.

8 The unresolved residual is a finite triad tape

For an exact K-supported Fourier–Galerkin path v, the retained equations cancel the projected part of (15). The nonlinear product can only generate modes in the finite $2K$ cube. For omitted output mode q,

$$\widehat{r}_q = P_q \left[i \sum_{p+s=q} (s \cdot \widehat{v}_p) \widehat{v}_s \right], \qquad \|q\|_\infty > K, \tag{19}$$

and

$$\|r\|_{H^{-1}}^2 = \sum_{q \neq 0} \frac{|\widehat{r}_q|^2}{|q|^2}. \tag{20}$$

For fixed K these are finite sums. Likewise, a conservative Fourier-polynomial bound is

$$\|\nabla v\|_\infty \le \sum_k |k|\,|\widehat{v}_k|. \tag{21}$$

Thus the analytic adapter reduces the remaining end-to-end numerical work to certified time integration/enclosure of finite quantities.

9 Validated RK4 continuous-time enclosure: the remaining frontier

The current finite solver uses explicit RK4 at discrete nodes. Nodewise residual samples do not certify the integrals in (16). The remaining numerical obligation is therefore precise: construct from the floating-point RK4 tape a validated continuous-time finite Fourier path $v_h(t)$ and computable enclosures

$$A_T \le \overline{A}_T, \qquad B_T \le \overline{B}_T. \tag{22}$$

Once (22) is available together with a certified initial mismatch, the relative-energy theorem gives the end-to-end bound

$$\|Q_K u(T)\|_2 \le e^{\overline{A}_T/2}\left(e_0^2 + \frac{\overline{B}_T}{\nu}\right)^{1/2}. \tag{23}$$

A practical validated implementation can use a piecewise-polynomial continuous extension of each RK4 step, outward-rounded interval bounds for its coefficients, interval evaluation of the finite triad residual on each time cell, and certified quadrature/enclosure of the resulting nonnegative polynomial or interval majorant. This paper does not claim that this final interval-enclosure layer has already been completed. In the accompanying Toledo registry it remains the explicit open target `PROP-EPSC-15`.

10 Executable finite evidence

The analytic claims above are deliberately separated from finite numerical diagnostics. The accompanying reproduction system currently records the following finite checks.

Check	Recorded result	Tier
Analytic shear finite recurrence	max error 3.886×10^{-16}	finite diagnostic
Direct triad vs padded FFT, $K = 2$	RHS diff 1.041×10^{-16}	finite diagnostic
One RK4 step, direct vs FFT	diff 8.674×10^{-19}	finite diagnostic
Taylor–Green nested gate, $K = 4$	$\delta_K = 3.551 \times 10^{-8}$	finite diagnostic
Taylor–Green nested gate, $K = 5$	$\delta_K = 7.741 \times 10^{-10}$	finite diagnostic
Spectral tail algebra	PASS on deterministic finite spectra	finite diagnostic
Terminal no-go witness	PASS; high-mode amplitude scalable	finite diagnostic
Energy-budget finite algebra	PASS	finite diagnostic
Relative-energy finite helper algebra	PASS	finite diagnostic

Table 1: Finite executable evidence. These checks validate finite algebra or recorded numerical instances; they do not upgrade the analytic theorems to machine proofs and do not certify turbulent DNS adequacy.

The short Taylor–Green nested-refinement run used $\nu = 0.01$, $\Delta t = 0.005$, and $T = 0.05$. The operational nested diagnostic passed at $K = 4$ and was confirmed at $K = 5$ under its declared thresholds. This does not imply that $K = 5$ contains all continuum information. A separate external-reference test at coarse turbulent resolution previously found that $K \leq 3$ did not reproduce the published $Re = 1600$ Taylor–Green dissipation curve; the present short-horizon stability check does not overturn that result.

11 Repository architecture and provenance

The research is intentionally split across four public repositories so that interpretation, provenance, executable mathematics, and domain-specific evidence cannot silently collapse into one another:

- `information-discrete-math`: reusable EPSC algorithms and certificate helpers;
- `toledo`: proposal/equation lineage and verification status;
- `readout-problem-navier-stokes`: Navier–Stokes analysis, manuscript, finite evidence, reproduction ledger, and claim boundary;
- `readout_genesis`: interpretation/application map only; NS results are not promoted to root ontology.

The EPSC lineage is registered as `PROP-EPSC-01` through `PROP-EPSC-15`. Classical Fourier, energy, Sobolev, Leray–Hopf and relative-energy inequalities are registered for lineage and role, not claimed as original mathematics.

AI systems were used for drafting, code integration, adversarial consistency checking, and reproducibility work. The human author is responsible for the mathematical claims, source selection, interpretation, and final release decisions. AI assistance is not treated as authorship or as certification of truth.

12 Claim boundary

The strongest positive statement in this paper is target-specific. A rigorous omitted-tail certificate exists for Leray–Hopf trajectories in $L_t^2 L_x^2$, and conditional/a-posteriori terminal certificates exist from declared richer information. A residual-based adapter converts a certified smooth comparison path into a terminal L^2 bound. The current numerical solver is not yet an end-to-end continuum certificate because the validated continuous-time enclosure of its floating-point RK4 tape remains open.

In particular, this work does *not* establish global smoothness, finite-time singularity, uniqueness of Leray–Hopf weak solutions, an autonomous finite closure of three-dimensional Navier–Stokes, universal computational speedup, or turbulent DNS accuracy at coarse cutoff. It does not solve or partially solve the Clay Millennium existence-and-smoothness problem. Its role is narrower: it turns an ambiguous statement that a finite calculation "looks converged" into a set of explicit mathematical obligations that can either be certified or fail closed.

13 Conclusion

The finite-to-infinite gap in Navier–Stokes computation is not one question. It depends on what is being read, in which norm, from which retained record, under which assumptions, and through which adapter. Terminal low-mode coefficients alone are non-identifying on an unrestricted class. By contrast, the Leray–Hopf energy inequality yields an explicit $O(K^{-1})$ spacetime certificate; a richer energy/dissipation tape yields a terminal budget bound; and a residual-based relative-energy estimate turns a certified finite comparison path into a terminal continuum bound. The remaining implementation frontier is now finite and concrete: validated continuous-time residual enclosure of the RK4 tape. This separation is the central practical result of the EPSC programme.

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